

Active Brownian particles

• 2D Active Brownian particles

$$\bullet \text{ Particle } i \text{ follows } \begin{cases} \mathrm{d}X_t^i &= \text{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

$$\bullet \quad \text{Pe} \geq 0, \sigma_x \geq 0$$

- $v(\theta) = (\cos(\theta), \sin(\theta))^T$ (or, more general, $v \in S^{d-1}$, ...)

- W_t^i 2D Brownian motion, B_t^i 1D Brownian motion

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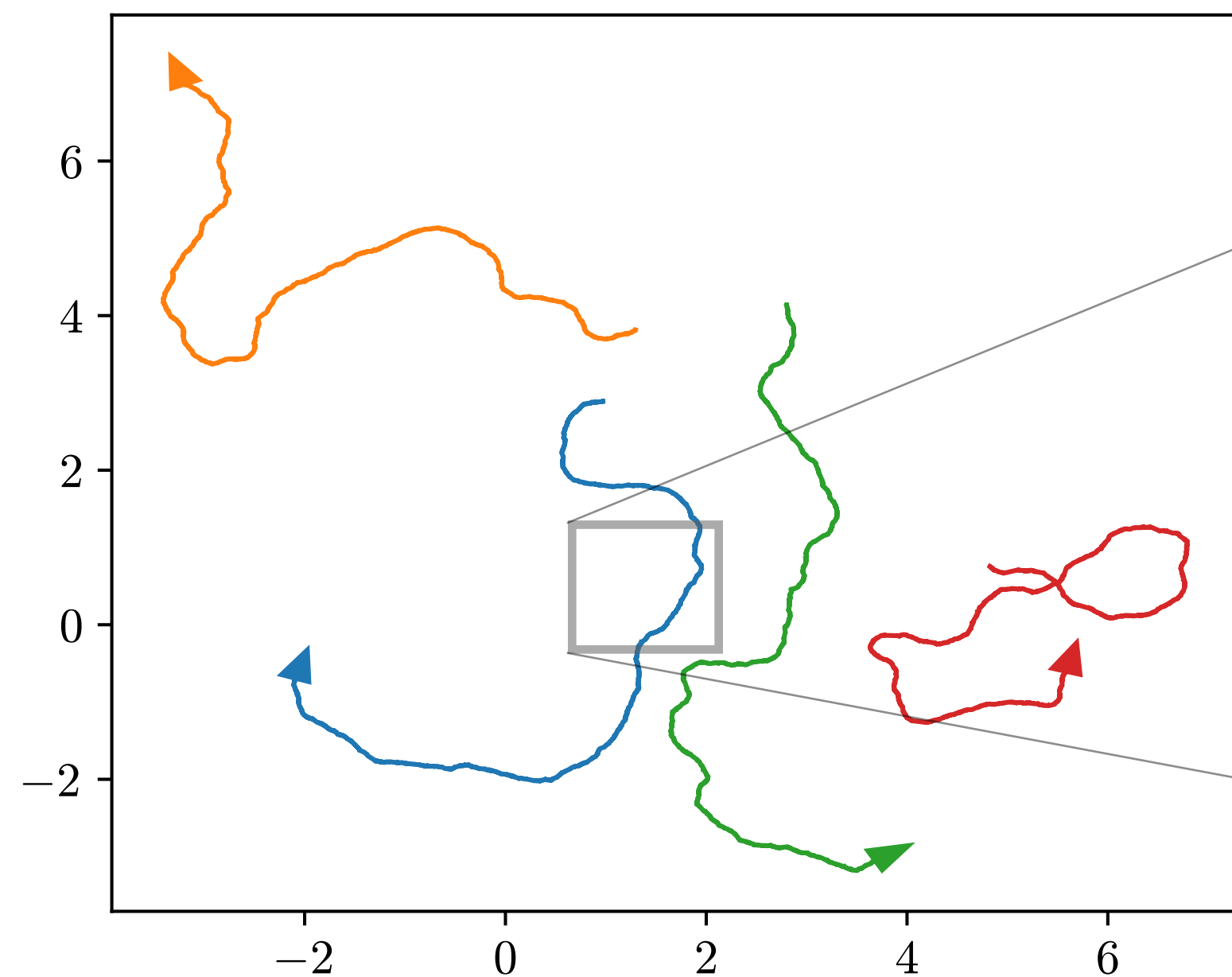
- Particle i follows
$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \sqrt{2} dB_t^i \end{cases}$$

- $\text{Pe} > 0, \sigma_x \geq 0$
- $v(\theta) = (\cos(\theta), \sin(\theta))^T$ (or, more general, $v \in \mathbb{S}^{d-1}, \dots$)
- W_t^i 2D Brownian motion, B_t^i 1D Brownian motion

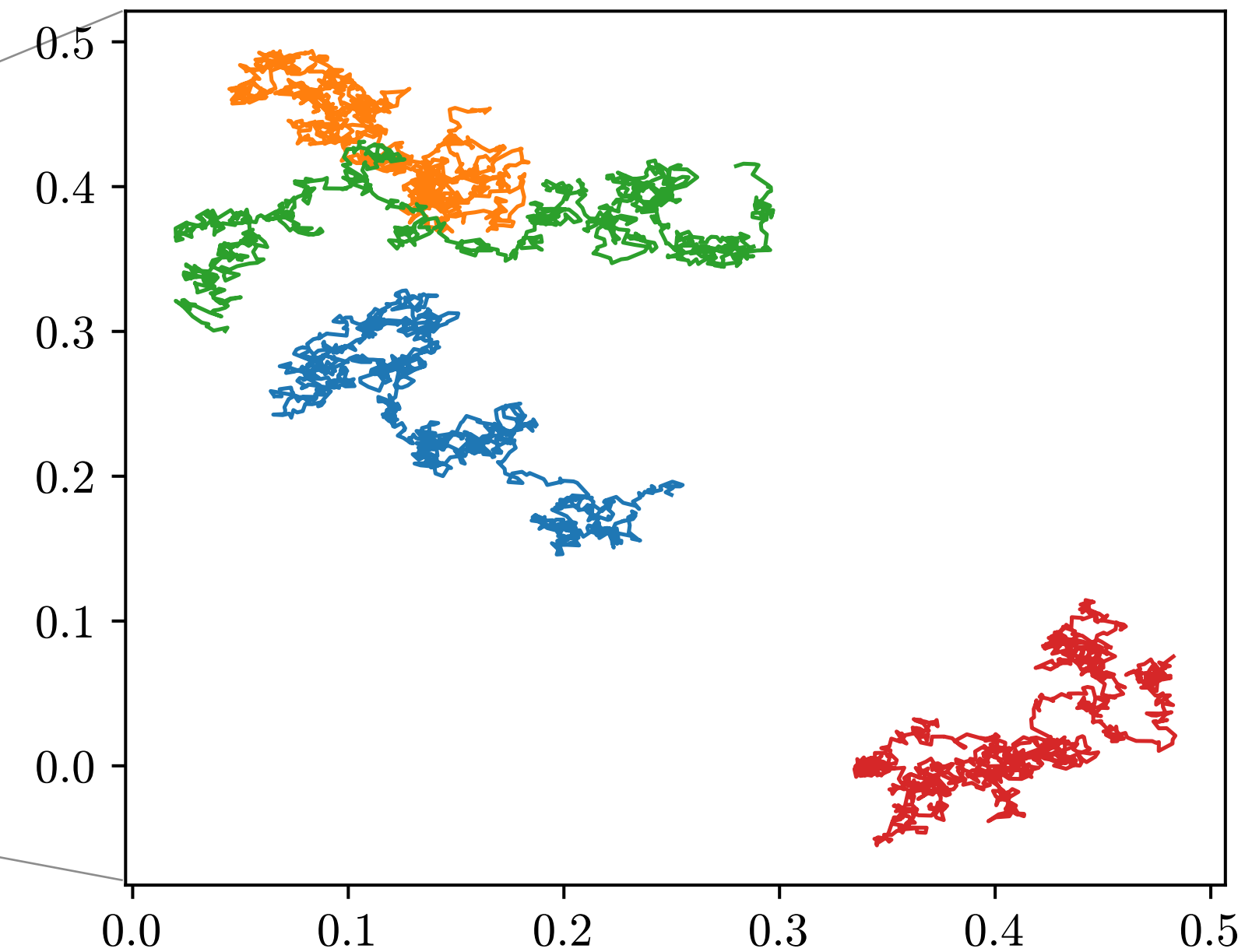
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$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \sqrt{2} dB_t^i \end{cases}$$



$\text{Pe} = 2, \sigma_x = 10^{-3}$



$\text{Pe} = 0, \sigma_x = 10^{-3}$